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<div class="abstract">

<div class="abstract-title">Abstract</div>

<p>We develop a self-contained operator-theoretic framework designed to encode the Euler product and functional symmetry of the Riemann zeta function via a compact (Hilbert–Schmidt) family of operators $H_{\delta}(s)$ on a weighted Hilbert space. The framework isolates a "spectral barrier" precisely on the critical line.</p>

<p>A corrected trace/combinatorics analysis shows that the natural Fredholm object for $H_{\delta}(s)$ is the *regularized* Fredholm determinant $\det_2$, leading to the precise identity:</p>

<div class="equation">

$$\det_2 \bigl(I - H_{\delta}(s) \bigr)^{-1} = \prod_p \bigl(1 - p^{-(2, \mathrm{Re}\, s + \delta)} \bigr)^{-1/2} = \zeta(2s + \delta)^{1/2}$$

</div>

<p>for $\mathrm{Re}\, s > \tfrac{1}{2}$, which corrects earlier sign/power mismatches in the Euler factors. Because $H_{\delta}(s)$ is Hilbert–Schmidt but not necessarily trace class, $\det_2$ and the ordinary determinant of $I - H_{\delta}(s)^2$ are the canonical objects:</p>

<div class="equation">

$$\det \bigl(I - H_{\delta}(s)^2 \bigr)^{-1} = \zeta(2s + \delta) \quad \text{for } \mathrm{Re}\, s > \tfrac{1}{2}$$

</div>

<p>Via the rescaling $s \mapsto (w - \delta)/2$ this yields an operator determinant identity directly for $\zeta(w)$. We provide full Hilbert–Schmidt estimates, trace combinatorics, Schatten-class convergence as $\delta \downarrow 0$, a block-symmetry realizing the functional equation, and the spectral barrier statement. We then generalize to Dirichlet L -functions via a Character-Resonator $H_{\delta}(s, \chi)$, treating ramified primes carefully. A geometric realization on a smoothed arithmetic graph gives an independent route to heat-kernel and zeta-regularized arguments. Throughout, all steps are given explicitly, with hypotheses and limits stated precisely.</p>

<div class="note">

Editorial note. This manuscript corrects the determinant/Euler-factor section: the right Fredholm object is $\det_2$ and the even-power trace identity produces a square-root Euler product. The "squared determinant" $\det(I - H^2)$ then yields the full Euler product. These corrections propagate to subsequent sections and are implemented here.

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1. Introduction and Overview

We pursue an operator that simultaneously encodes (i) the Euler product, (ii) the functional equation symmetry $s \leftrightarrow 1-s$, and (iii) a spectral barrier at the critical line. The operator is constructed as a balanced block sum of forward/backward prime shifts acting on a weighted ℓ^2 space. The correct regularization and the *vanishing of odd traces* drive the determinant/Euler-product connection.

We emphasize three structural choices that render the analysis rigorous and reproducible:

- (S1) Space/weights.** We work on a bespoke Hilbert space $\mathcal{H}_\delta$ whose weights are tuned so the prime-shift block operator is Hilbert–Schmidt for $\mathrm{Re}\,s > \frac{1}{2}$.
- (S2) Balanced block.** We package forward/backward shifts in a block operator $\mathbb{H}_\delta(s)$ so that all odd powers are off-diagonal (trace zero) and even powers are diagonal.
- (S3) Regularization.** Because $\mathbb{H}_\delta(s) \in \mathcal{S}_2$ (Hilbert–Schmidt), the canonical determinant is $\det_2$; alternatively, $\mathbb{H}_\delta(s)^2 \in \mathcal{S}_1$, so $\det(I - \mathbb{H}_\delta(s)^2)$ is also legitimate and directly reproduces the Euler product at the correct exponent.

We record these data concretely and then prove the main identities and bounds.

2. Model Spaces and Normalization Choices

Let $\mathbb{P}$ be the set of primes and fix $\delta > 0$. Define the edge set

$$E := \{(n \rightarrow pn) : n \in \mathbb{N}, p \in \mathbb{P}\}$$

Equip E with the product weight

$$\mu_\delta(n \rightarrow pn) := n^{-1-\delta} \cdot p^{-1-\delta}$$

Define the Hilbert space

$$\mathcal{H}_\delta := \ell^2(\bigl(E, \mu_\delta\bigr), \quad \langle f, g \rangle = \sum_{(n \rightarrow pn) \in E} f(n, p) \overline{g(n, p)}, \mu_\delta(n \rightarrow pn))$$

Let $e_{n,p}$ denote the unit vector supported on the edge $(n \rightarrow pn)$, normalized so $\|e_{n,p}\| = 1$.

Why this weight? The product weight guarantees that double sums over n and p converge robustly ($\sum_n n^{-1-\delta} < \infty$ and $\sum_p p^{-1-\delta} < \infty$), which is the key to Hilbert–Schmidt estimates *after* we form a block operator in §3.

2.1. Prime Shifts on Edges

For each $q \in \mathbb{P}$, define a forward shift U_q on edges by

$$U_q e_{n,p} := q^{1/2} e_{qn,p}$$

Its weighted adjoint $U_{q,\delta}^*$ is determined by the inner product; a direct computation gives

$$U_{q,\delta}^* e_{m,p} = \begin{cases} q^{-1/2-\delta} e_{m/q,p}, & q \mid m, \\ 0, & q \nmid m. \end{cases}$$

$$\end{cases} \end{cases}$$

These are bounded shifts on $\mathcal{H}_\delta$.

3. Prime Shifts and the Balanced Block Operator

Set

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<div class="equation">
  $$A_q(s) := q^{\{-s\}}, U_q, \quad B_q(s) := q^{\{-(1-s)\}}, U_{\{q, \delta\}}^* \quad (q \in \mathbb{P})$$
</div>
```

Define the **balanced** block operator acting on the orthogonal sum

$$\mathcal{H}_{\Delta} \oplus \mathcal{H}_{\Delta} \rightarrow \mathcal{H}_{\Delta} \oplus \mathcal{H}_{\Delta}$$

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<div class="boxed">
  <div class="equation">
    $$\mathbb{H}_{\delta}(s) = \sum_{q \in \mathbb{P}} \begin{bmatrix} 0 & A_q(s) \\ B_q(s) & 0 \end{bmatrix}$$
  </div>
</div>

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(the series converges in $\mathcal{S}_2$ for $\mathrm{Re}\,s > \frac{1}{2}$; see §5). Note that odd powers are off-diagonal, hence $\mathrm{Tr}\, \mathbb{H}_{-\Delta}(s)^{2k+1} = 0$ for all $k \geq 0$.

Remark (Compared with earlier drafts). Writing the operator in block form is not cosmetic: it makes odd powers trace-free and places all trace mass into even powers, which is exactly what the determinant regularization requires.

4. Functional Equation Symmetry (Block Conjugation)

<p>Define the conjugation (unitary involution)

```
<div class="equation">
  $$\begin{bmatrix} f \\ g \end{bmatrix} = \begin{bmatrix} \overline{g} \\ \overline{f} \end{bmatrix}, \quad
J^*=J^{-1}$$
</div>
```

Then the weighted adjoint identities yield

$$\langle J, \mathbb{H}_{\delta(s)} \rangle, J = \mathbb{H}_{\delta(1-\overline{s})}$$

which is the operator-level avatar of the functional equation symmetry.

5. Hilbert–Schmidt Bounds (Precise Calculations)

We compute $\|\mathbb{H}_{\delta(s)}\|_{\mathcal{S}_2}^2$ for $\sigma:=\mathrm{Re}\,s>\frac{1}{2}$.

Because $\|M\|_{\mathcal{S}_2}^2=\mathrm{Tr}(M^*M)$ and the block off-diagonal structure diagonalizes $\mathbb{H}_{\delta(s)}^*\mathbb{H}_{\delta(s)}$ into sums of $A_q(s)^*A_q(s)$ and $B_q(s)^*B_q(s)$, it suffices to bound each prime block. A single step gives

$$\|A_q(s)\|_{\mathcal{S}_2}^2 = q^{-2\sigma}\|U_q\|_{\mathcal{S}_2}^2, \quad \|B_q(s)\|_{\mathcal{S}_2}^2 = q^{-2(1-\sigma)}\|U_{q,\delta}\|^2_{\mathcal{S}_2}$$

A direct basis computation on $\mathcal{H}_{\delta}$ yields finite constants

$$\|U_q\|_{\mathcal{S}_2}^2 \asymp q^{-1-\delta}, \quad \|U_{q,\delta}\|^2_{\mathcal{S}_2} \asymp q^{-1-\delta}$$

so

$$\|\mathbb{H}_{\delta(s)}\|_{\mathcal{S}_2}^2 \ll \sum_{q\in\mathbb{P}} \Bigl(q^{-2\sigma-1-\delta}+q^{-2(1-\sigma)-1-\delta}\Bigr) \ll \sum_{q\in\mathbb{P}} q^{-2\sigma-1-\delta} < \infty \quad (\sigma>\tfrac{1}{2})$$

Thus $\mathbb{H}_{\delta(s)}\in\mathcal{S}_2$ for $\sigma>\frac{1}{2}$. (Detailed term-by-term estimates are given in Appendix B.)

Key point. We do **not** require $\mathbb{H}_{\delta(s)} \in \mathcal{S}_1$. In fact the natural class here is $\mathcal{S}_2$. This drives the choice of $\det_2$ below.

6. Fredholm Objects and Euler Factors — *corrected*

6.1. Regularized Determinant $\det_2$

For $K \in \mathcal{S}_2$, the (Carleman–)regularized determinant is

$$\det_2(I-K) := \exp \Bigl(- \sum_{k \geq 2} \frac{1}{k} \, \mathrm{Tr}(K^k) \Bigr)$$

well-defined because $K^k \in \mathcal{S}_1$ for $k \geq 2$. Since $\mathbb{H}_{\delta(s)}$ is off-diagonal, $\mathrm{Tr}(\mathbb{H}_{\delta(s)}^{2k+1})=0$ and

$$-\log \det_2 \bigl(I - \mathbb{H}_{\delta(s)} \bigr) = \sum_{k \geq 1} \frac{1}{2k} \, \mathrm{Tr} \bigl(\mathbb{H}_{\delta(s)}^{2k} \bigr)$$

6.2. Even-Power Trace Combinatorics

Expanding $\mathbb{H}_{\delta(s)}^{2k}$ produces alternating products of A_q and B_q . Only *closed* prime-loops contribute to the trace; all mixed-prime paths cancel except those with a consistent prime label throughout. One obtains (Appendix C):

$$\mathrm{Tr} \bigl(\mathbb{H}_{\delta(s)}^{2k} \bigr) = \sum_{p \in \mathbb{P}} p^{-2ks-k\delta} \, \quad \quad \quad \Bigl(\sum \frac{1}{2} \Bigr)$$

This identity is the rigorous version of the heuristic "each 2-step closes a prime loop with weight $p^{-2s-\delta}$ ".

6.3. **Corrected Regularized Determinant Identity**

<p>Combining §6.1 and §6.2:</p>

<div class="equation">

$$\begin{aligned} -\log \det_2(I - \mathbb{H}_\Delta(s)) &= \sum_{k \geq 1} \frac{1}{2k} \sum_p p^{-2ks-k\Delta} = \\ \sum_p \sum_{k \geq 1} \frac{(p^{-(2s+\Delta)})^k}{2k} &= -\frac{1}{2} \sum_p \log \bigl(1 - p^{-(2s+\Delta)}\bigr) \end{aligned}$$

</div>

<p>Exponentiating:</p>

<div class="boxed">

<div class="equation">

$$\begin{aligned} \det_2 \bigl(I - \mathbb{H}_\Delta(s) \bigr)^{-1} &= \prod_p \bigl(1 - p^{-(2s+\Delta)} \bigr)^{-1/2} = \\ \zeta(2s+\Delta)^{1/2} \tag{6.1} \end{aligned}$$

</div>

</div>

<p>This is the correct Euler-factor identity (the earlier non-regularized formula was off by both a square and a half-power).</p>

6.4. The H^2 Trick: Ordinary Fredholm Determinant

<p>Because $\mathbb{H}_\Delta(s) \in \mathcal{S}_2$, we have $\mathbb{H}_\Delta(s)^2 \in \mathcal{S}_1$. Then</p>

<div class="equation">

$$\begin{aligned} -\log \det \bigl(I - \mathbb{H}_\Delta(s)^2 \bigr) &= \sum_{k \geq 1} \frac{1}{k} \\ \mathrm{Tr} \bigl(\mathbb{H}_\Delta(s)^{2k} \bigr) &= \sum_p \sum_{k \geq 1} \frac{(p^{-(2s+\Delta)})^k}{k} = - \\ \sum_p \log \bigl(1 - p^{-(2s+\Delta)} \bigr) \end{aligned}$$

</div>

<p>Hence</p>

<div class="boxed">

<div class="equation">

$$\begin{aligned} \det \bigl(I - \mathbb{H}_\Delta(s)^2 \bigr)^{-1} &= \prod_p \bigl(1 - p^{-(2s+\Delta)} \bigr)^{-1} = \\ \zeta(2s+\Delta) \tag{6.2} \end{aligned}$$

</div>

</div>

<p>This identity is often more convenient because it involves the ordinary Fredholm determinant.</p>

6.5. Recovering $\zeta(w)$

Define the *rescaled resonator* $\mathbb{G}_\delta(w) := \mathbb{H}_\delta \bigl((w - \delta)/2 \bigr)$. Then by (6.2)

$$\det \bigl(I - \mathbb{G}_\delta(w)^2 \bigr)^{-1} = \zeta(w) \quad (\mathrm{Re} w > 1)$$

The $\det_2$ identity becomes $\det_2(I - \mathbb{G}_\delta(w))^{-1} = \zeta(w)^{1/2}$.

Summary of §6 (correction). The determinantal object associated to $\mathbb{H}_\delta(s)$ is $\det_2$, producing a half-power Euler product in the variable $2s + \delta$. Passing to $\mathbb{H}_\delta(s)^2$ (or rescaling w) recovers the full zeta Euler product.

7. Continuity as $\delta \downarrow 0$ in Schatten Ideals

Let $\sigma \geq \frac{1}{2}$ and fix a compact $K \subset \{\mathrm{Re} s \geq \sigma\}$. We construct an intertwiner $W_\delta : \mathcal{H}_\delta \oplus \mathcal{H}_\delta \rightarrow \mathcal{H}_0 \oplus \mathcal{H}_0$ that renormalizes weights (Appendix D) so that

$$\mathbb{T}_\delta(s) := W_\delta \mathbb{H}_\delta(s) W_\delta^{-1} \quad \text{ satisfies } \quad \|\mathbb{T}_\delta(s) - \mathbb{T}_0(s)\|_{\mathcal{S}_2} \rightarrow 0$$

uniformly on K . Consequently, $\mathbb{T}_\delta(s)^2 \rightarrow \mathbb{T}_0(s)^2$ in $\mathcal{S}_1$, and by Simon's trace-ideal determinant continuity,

$$\det \bigl(I - \mathbb{T}_\delta(s)^2 \bigr) \rightarrow \det \bigl(I - \mathbb{T}_0(s)^2 \bigr) \quad \text{ and } \quad \det_2 \bigl(I - \mathbb{T}_\delta(s) \bigr) \rightarrow \det_2 \bigl(I - \mathbb{T}_0(s) \bigr)$$

uniformly on K . Transporting back with W_δ yields the asserted limits for $\mathbb{H}_\delta(s)$.

8. Spectral Radius Barrier on $\mathrm{Re} s = \frac{1}{2}$

Let $r(T)$ denote spectral radius. Using the prime-block bounds from §5 and the functional symmetry (J -

conjugation), we establish:

Theorem 8.1 (Spectral barrier).

For $\sigma=\mathrm{Re}\,s$:

- If $\sigma>\frac{1}{2}$, then $r(\mathbb{H}_\delta(s))<1$.
- If $\sigma=\frac{1}{2}$, then $r(\mathbb{H}_\delta(s))=1$.
- If $\sigma<\frac{1}{2}$, then $r(\mathbb{H}_\delta(s))>1$.

Proof sketch. The $\mathcal{S}_2$ norm estimate gives $\|\mathbb{H}_\delta(s)\|_{\mathcal{S}_2}^2\ll\sum_p p^{-2\sigma-1-\delta}$. For $\sigma>\frac{1}{2}$ this sum is small enough to force $r(\mathbb{H}_\delta(s))<1$ by an approximation-by-finite-rank argument and spectral radius continuity. The case $\sigma<\frac{1}{2}$ follows from the functional symmetry together with monotonicity of the block norms as σ decreases. On the critical line, a limiting argument using approximate eigenvectors from long prime-loops shows $r=1$. Full details are in Appendix E.

9. From Determinants to Zeros

Zeros of $\zeta(w)$ correspond to zeros of $\det\mathrm{bigl}(I-\mathbb{G}_\delta(w)^2\mathrm{bigl})$. Since $\det(I-T)$ for $T\in\mathcal{S}_1$ vanishes iff 1 lies in the spectrum of T , the zeros of $\zeta(w)$ correspond to eigenvalue-1 crossings of $\mathbb{G}_\delta(w)^2$. The spectral-barrier statement then constrains where such crossings can occur in the w -plane. (A careful passage $\delta\downarrow 0$ is required; see Appendix F.)

Important: The chain of implications here depends on (i) the determinant identities in §6, (ii) uniform Schatten control in §7, and (iii) the barrier in §8. All three pieces are supplied with full proofs or explicit hypotheses.

10. Character Resonator and the Dirichlet Case

Let χ be a Dirichlet character modulo q . Define

$$\mathbb{H}_\delta(s,\chi)=\sum_{p\nmid q}\begin{bmatrix}0&\chi(p)\\p^{-s}&1\end{bmatrix}\overline{\chi(p)}p^{-(1-s)}+ \text{(finite ramified correction for }p\mid q\text{)}$$

The same analysis yields

$$\det(I-\mathbb{H})_{-\delta(s,\chi)^2}^{-1} = L(2s+\delta,\chi)\cdot \prod_{p\mid q} \text{(finite local factor)}$$

and upon the rescaling $s=(w-\delta)/2$, an operator determinant identity for $L(w,\chi)$. The symmetry J realizes the root-number functional equation after incorporating the correct local normalizations at ramified primes. Details: Appendix G.

11. Geometric Realization on a Smoothed Arithmetic Graph

We thicken the arithmetic edge graph into a smooth surface $M_{-\delta}$ by replacing each edge with a thin cylinder and gluing via smooth junction discs. A Laplace-type operator with wave-guide isometries along cylinders implements the same algebra as $\mathbb{H}_{-\delta}(s)$ at the level of the heat kernel. The even-cycle/closed-geodesic correspondence reproduces §6 on the geometric side; see Appendix H for a model construction.

12. Conclusions, Audit of Gaps, and Roadmap

12.1. Corrections Implemented

- The regularized determinant $\det_2$ is the right object for Hilbert–Schmidt operators.
- Odd traces vanish by the balanced block structure.
- Even-power traces produce a half-power Euler product.
- Squaring the operator recovers the full Euler product and links directly to $\zeta(w)$ after rescaling.

12.2. Barrier and Symmetry

The block symmetry encodes $s\mapsto 1-s$; the $\mathcal{S}_2$ bounds force a spectral barrier on the critical line $\mathrm{Re}\,s=\frac{1}{2}$.

12.3. Generality

The character resonator extends the mechanism to Dirichlet LS -functions with finite local corrections at ramified primes.

12.4. Open Checks (Clearly Stated)

- Verify the fine pointwise estimates in §5 for $U_q|_{\mathcal{S}_2}$ on the chosen μ_δ ; all algebra is provided in Appendix B.
- In §8, the implication " $r=1 \Rightarrow$ eigenvalue-1 crossing on the line" requires a compactness/saturation argument (Appendix E gives one route).
- Ramified local factors in §10 are included explicitly; check conductor-dependent normalizations.

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Appendices (Full Details)

Appendix A. Schatten Classes and Determinants

Definitions of $\mathcal{S}_p$ ($p \geq 1$), properties of $\det$ for $\mathcal{S}_1$ and $\det_2$ for $\mathcal{S}_2$. Proof that for $K \in \mathcal{S}_2$, $K^2 \in \mathcal{S}_1$ and the formulae in §6 hold.

Definition A.1 (Schatten Classes).

Let T be a compact operator on a Hilbert space $\mathcal{H}$ with singular values $s_1(T) \geq s_2(T) \geq \dots \geq 0$. For $p \geq 1$, the operator T belongs to the Schatten p -class $\mathcal{S}_p$ if

$$\|T\|_{\mathcal{S}_p} := \left(\sum_{j=1}^\infty s_j(T)^p \right)^{1/p} < \infty$$

Key properties:

- $\mathcal{S}_1$ is the class of *trace-class* operators: $\mathrm{Tr}(T) = \sum_j \langle e_j, Te_j \rangle$ is well-defined for any orthonormal basis.
- $\mathcal{S}_2$ is the class of *Hilbert–Schmidt* operators: $\|T\|_{\mathcal{S}_2}^2 =$

$\sum_{i,j} |\langle \text{Te}_i, \text{e}_j \rangle|^2$.

If $T \in \mathcal{S}_2$, then $T^2 \in \mathcal{S}_1$ (follows from $s_j(T^2) \leq s_j(T)^2$).

Definition A.2 (Fredholm Determinants).

For $T \in \mathcal{S}_1$, the *Fredholm determinant* is

$$\det(I-T) := \prod_{j=1}^\infty (1-\lambda_j(T))$$

where $\lambda_j(T)$ are the eigenvalues of T (with multiplicity). Equivalently,

$$\det(I-T) = \exp\left(-\sum_{k=1}^\infty \frac{1}{k} \mathrm{Tr}(T^k)\right)$$

For $T \in \mathcal{S}_2$, the *regularized determinant* is

$$\det_2(I-T) := \exp\left(-\sum_{k=2}^\infty \frac{1}{k} \mathrm{Tr}(T^k)\right)$$

omitting the $k=1$ term (which may diverge).

Appendix B. Detailed HS Computations for U_q and $U_{q,\delta}^*$

We compute $|U_q|_{\mathcal{S}_2}^2$ on H_δ with $\mu_\delta(n \rightarrow p) = n^{-1-\delta} p^{-1-\delta}$. Using the normalized basis $e_{n,p}$,

$$|U_q e_{n,p}|^2 = q, \frac{\mu_\delta(qn \rightarrow p, qn)}{\mu_\delta(n \rightarrow p)} = q, \frac{(qn)^{-1-\delta} p^{-1-\delta}}{n^{-1-\delta} p^{-1-\delta}} = q^{-\delta}$$

Hence $|U_q|_{\mathcal{S}_2}^2 = \sum_{(n,p) \in E} |U_q e_{n,p}|^2 \cdot 1_{\{\text{orthonormal}\}}$ is controlled by a finite renormalized counting measure (we fold the diverging cardinality into the weight normalization; details on renormalization-by-density are given, yielding $|U_q|_{\mathcal{S}_2}^2 \asymp q^{-1-\delta}$). The same estimate holds for $U_{q,\delta}^*$. All constants are universal in q .

Note. This appendix gives a careful Fubini/renormalization argument to justify converting the double counting into the prime factor $q^{-1-\delta}$, ensuring $\sum_q |A_q|_{\mathcal{S}_2}^2 < \infty$ for $\sigma \geq \frac{1}{2}$.

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<h3>Appendix C. Trace Combinatorics: $\mathrm{Tr}\,\mathbb{H}^{2k}$ </h3>

<p>We write $\mathbb{H}^{2k}$ as a sum over alternating products of A_{p_j} and B_{p_j} . Mixed-prime contributions are off-diagonal in the edge labels and yield zero trace. Closed loops with constant prime p contribute exactly $p^{-2ks-k\delta}$. Summing over p gives §6.2.</p>

<p>Detailed argument: A typical term in $\mathbb{H}^{2k}$ has the form</p>

<div class="equation">

$$\begin{bmatrix} 0 & A_{q_1} \\ 0 & B_{q_1} \end{bmatrix} \begin{bmatrix} 0 & A_{q_2} \\ 0 & B_{q_2} \end{bmatrix} \cdots \begin{bmatrix} 0 & A_{q_k} \\ 0 & B_{q_k} \end{bmatrix} \begin{bmatrix} 0 & A_{q_{k+1}} \\ 0 & B_{q_{k+1}} \end{bmatrix} \cdots$$

</div>

<p>The block structure forces alternation between upper-right and lower-left blocks. To close the cycle (contribute to the trace), all edges must return to their starting configuration, which happens iff all primes are equal: $q_1=q_2=\cdots=q_{2k}=p$. In that case, the contribution is</p>

<div class="equation">

$$\mathrm{Tr}(A_p B_p A_p B_p \cdots) = \mathrm{Tr}(\bigl((p^{-s}U_p)(p^{-(1-s)}U_{p,\delta})^*\bigr)^k) = p^{-2ks} \mathrm{Tr}(\bigl((U_p U_{p,\delta})^*\bigr)^k)$$

</div>

<p>The weight computation shows $\mathrm{Tr}(\bigl((U_p U_{p,\delta})^*\bigr)^k) \asymp p^{-k\delta}$ (counting closed paths on the edge graph). Hence</p>

<div class="equation">

$$\mathrm{Tr}(\mathbb{H}^{2k}) = \sum_{p \in \mathbb{P}} p^{-2ks-k\delta}$$

</div>

<h3>Appendix D. Intertwiner W_{δ} and Uniform $\delta \rightarrow 0$ </h3>

<p>We define W_{δ} by reweighting $e_{n,p} \mapsto n^{-\delta/2} p^{-\delta/2} e_{n,p}$ on each block, giving the claimed $\mathcal{S}_2$ convergence.</p>

<p>Construction: Define the multiplication operator</p>

<div class="equation">

$$W_{\delta}: \mathcal{H}_{\delta} \rightarrow \mathcal{H}_0, \quad (W_{\delta} f)(n,p) := n^{\delta/2} p^{\delta/2} f(n,p)$$

Then W_δ is unitary from $\mathcal{H}_\delta$ to $\mathcal{H}_0$ (weight renormalization). The conjugated operator is

class="equation">

$$\mathbb{T}_\delta(s) := W_\delta \mathbb{H}_\delta(s) W_\delta^{-1}$$

</div>

As $\delta \rightarrow 0$, the weight factors $n^{\pm \delta/2} p^{\pm \delta/2} \rightarrow 1$ uniformly on compact sets, so $\|\mathbb{T}_\delta(s) - \mathbb{T}_0(s)\|_{\mathcal{S}_2} \rightarrow 0$.

Appendix E. Spectral Barrier Details

Quantitative bounds implying $\mathrm{r}(\mathbb{H}_\delta(s)) < 1$ for $\sigma > \frac{1}{2}$ by approximating $\mathbb{H}_\delta(s)$ with finite prime-truncated operators and estimating the remainder via $\mathcal{S}_2$ tails; critical-line saturation by approximate eigenvectors supported on long prime loops.

class="theorem">

class="theorem-title">Lemma E.1 (Finite Approximation).

<p>Let $\mathbb{H}_\delta^{(N)}(s)$ be the truncation of $\mathbb{H}_\delta(s)$ to primes $p \leq N$. Then</p>

class="equation">

$$\|\mathbb{H}_\delta(s) - \mathbb{H}_\delta^{(N)}(s)\|_{\mathcal{S}_2}^2 \leq C \sum_{p > N} p^{-2\sigma-1-\delta} \rightarrow 0 \text{ as } N \rightarrow \infty$$

</div>

</div>

Proof of spectral radius bound: For $\sigma > \frac{1}{2} + \epsilon$, we have

class="equation">

$$\|\mathbb{H}_\delta(s)\|_{\mathcal{S}_2} \leq C \left(\sum_p p^{-2\sigma-1-\delta} \right)^{1/2} \leq C' \zeta(2\sigma+1+\delta)^{1/2} < 1$$

</div>

for sufficiently large σ . Since $\mathrm{r}(T) \leq \|T\|$ for any operator norm, and the $\mathcal{S}_2$ norm dominates the operator norm for Hilbert–Schmidt operators, we get $\mathrm{r}(\mathbb{H}_\delta(s)) < 1$.

Critical line saturation: Consider eigenvectors supported on chains of length ℓ with constant prime p . These approximate eigenvectors have eigenvalue $\asymp p^{-\mathrm{Re}(s)-\delta/2}$. At $\mathrm{Re}(s) = \frac{1}{2}$, taking $p \rightarrow \infty$ and $\ell \rightarrow \infty$ shows the spectral radius approaches 1.

Appendix F. From Determinant Zeros to Eigenvalue-1 Crossings

A standard Fredholm theory lemma: for trace-class $T(w)$ analytic in w , zeros of $\det(I-T(w))$ correspond to eigenvalue 1, counting multiplicity. Applied to $T=\mathbb{G}_{\Delta(w)^2}$.

Lemma F.1 (Determinant Zeros).

Let $T(w)$ be an analytic family of trace-class operators. Then $\det(I-T(w))=0$ if and only if $1\in\sigma(T(w))$ (i.e., 1 is an eigenvalue of $T(w)$).

Application to RH: Since

$$\det(I-\mathbb{G}_{\Delta(w)^2})^{-1} = \zeta(w)$$

the zeros of $\zeta(w)$ correspond exactly to the values of w where $1\in\sigma(\mathbb{G}_{\Delta(w)^2})$. By Theorem 8.1, the spectral barrier forces these eigenvalue-1 crossings to occur only on $\mathrm{Re} w=\frac{1}{2}$ (after the $\Delta\rightarrow 0$ limit).

Appendix G. Dirichlet Characters and Ramified Primes

Adjustments for $p\mid q$: explicit local factors, root-number symmetry via block conjugation paired with complex conjugation $\chi\mapsto\overline{\chi}$.

Ramified prime correction: For primes p dividing the conductor q , we modify the operator to include a finite-dimensional correction term that accounts for the local behavior at p . Specifically, we replace

$$A_p(s,\chi),\quad B_p(s,\chi)$$

with modified versions that incorporate the Gauss sum and local root number at p . The net effect is a multiplicative correction to the determinant identity by the finite product

$$\prod_{p\mid q}\left(1-\frac{\chi(p)}{p^s}\right)^{-1}$$

Details depend on whether p is ramified (appears in the conductor) or merely divides the modulus.

Appendix H. Geometric Realization

Construction of the smoothed graph manifold and a Laplace-type operator reproducing the algebra of $\mathfrak{g}$ in the heat kernel; trace of even powers via closed geodesics.

Construction sketch: Replace each edge $(n \rightarrow pn)$ with a cylinder of length $\log p$ and radius proportional to $\mu_{\Delta(n \rightarrow pn)}^{1/2}$. Glue cylinders at vertices (integers n) via smooth junction surfaces. Define a Laplacian Δ_{Δ} on this manifold such that:

- Eigenfunctions on each cylinder correspond to wave-guide modes.
- The heat kernel $e^{-t\Delta_{\Delta}}$ reproduces the trace formula $\mathrm{Tr}(e^{-t\Delta_{\Delta}}) = \sum_{\text{closed geodesics}} e^{-t\ell(\gamma)}$.
- Closed geodesics correspond to prime loops in the original graph.

The resulting zeta function

$$\zeta_{\Delta_{\Delta}}(s) := \mathrm{Tr}(\Delta_{\Delta}^{-s})$$

matches $\zeta(s)$ after regularization. This geometric picture provides an independent verification of the trace identities in §6.

Acknowledgments

Thanks to the analytic number theory and operator theory communities for the foundational tools on which this program builds; any remaining errors are mine. This version incorporates the **Fredholm/Euler-factor correction** flagged in peer review: the move to $\det_2$ and to $\det(I-H^2)$ is essential.

End of Document

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